

Reinforcement Learning (CS60077)

Autumn 2026-27

Course Details

- Theory Teacher
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- Lecture Hours
Wed(12:00-12:55), Thurs(11:00-11:55), Fri(09:00-09:55) at CSE-119
- Evaluation
TA Evaluation: Marks 40 (Tests, Assignments, Term Project)
Mid Semester: Marks 30
End Semester: Marks 30

Grading will be relative

Course Materials

- Slides will be uploaded in Moodle.

Please enroll as student in Moodle with password “[RL@2026A](#)”.

- Books:

Richard S. Sutton and Andrew G. Barto; [Reinforcement Learning: An Introduction](#); 2nd Edition, MIT Press, 2020. [TEXTBOOK]

- ** For Reference Slides/Materials, Visit the following Course Pages:

[Course by Dr. David Silver \(Deepmind and UCL, UK\)](#)

[Course by Dr. Emma Brunskill \(Stanford, USA\)](#)

Important Dates

Schedule of Tests and Examinations

Date	Event
September 22 – October 01, 2026	MidSem Examination

List of Holidays

Date	Event
26-Aug-26	Id-e-Milad
04-Sep-25	Janmastami

Pre-requisite

- Probability Theory
- Linear Algebra

Probability Theory

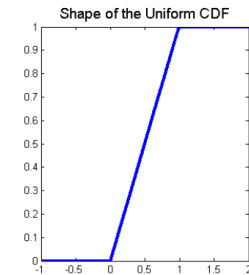
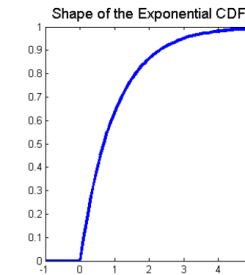
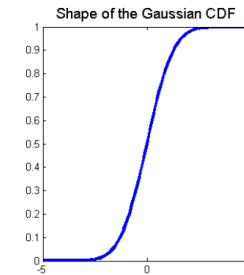
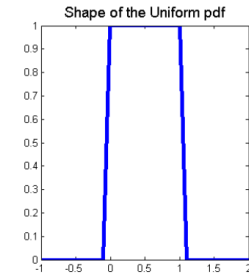
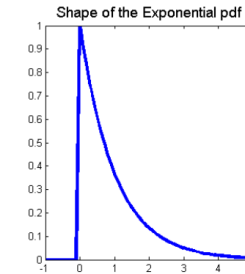
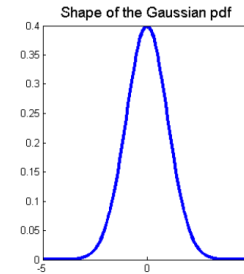
- Elements of Probability
 - Sample Space
 - Event Space or set of events
 - Probability measure

Probability Theory

- More on Probability
 - Conditional Probability
 - Independence

Probability Theory

- Random Variables
 - Cumulative distribution functions (CDF)
 - Probability mass functions (PMF)
 - Probability density functions (PDF)
 - Expectation
 - Variance



Probability Theory

- Two random variables
 - Joint and marginal distributions
 - Joint and marginal probability mass functions
 - Joint and marginal probability density functions
 - Conditional distributions
 - Bayes's rule
 - Independence
 - Expectation and covariance

Probability Theory

- Multiple random variables
 - We can define the **joint distribution function** of $X_1, X_2, \dots, X_n$, the **joint probability density function** of $X_1, X_2, \dots, X_n$, the **marginal probability density function** of X_1 , and the **conditional probability density function** of X_1 given $X_2, \dots, X_n$
 - Random vectors
 - Expectation
 - Covariance matrix
 - The multivariate Gaussian distribution

Linear Algebra

- Matrix

- The Identity Matrix and Diagonal Matrices
- The Transpose
- Symmetric Matrices
- The Trace
- Norms
- Linear Independence and Rank
- The Inverse
- Orthogonal Matrices
- Range and Nullspace of a Matrix
- The Determinant
- Quadratic Forms and Positive Semidefinite Matrices
- Eigenvalues and Eigenvectors
- Eigenvalues and Eigenvectors of Symmetric Matrices

Linear Algebra

- Matrix multiplication
 - Vector-vector
 - Matrix-Vector
 - Matrix-Matrix

Linear Algebra

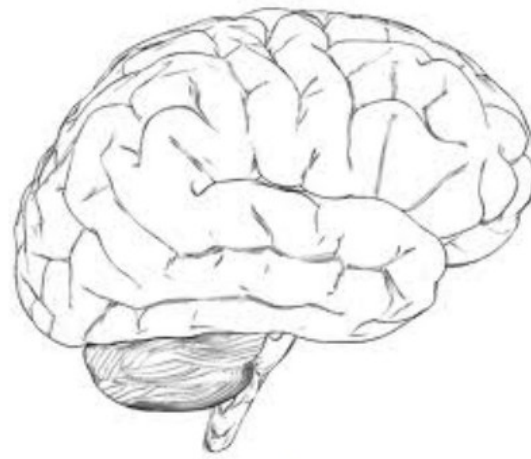
- Matrix Calculus
 - The Gradient
 - The Hessian
 - Gradients and Hessians of Quadratic and Linear Functions
 - Least Squares
 - Gradients of the Determinant
 - Eigenvalues as Optimization

A financial investment

- Goal: select actions to maximise total future reward
- Actions may have long term consequences
- Reward may be delayed
- It may be better to sacrifice immediate reward to gain more long-term reward

observation

O_t



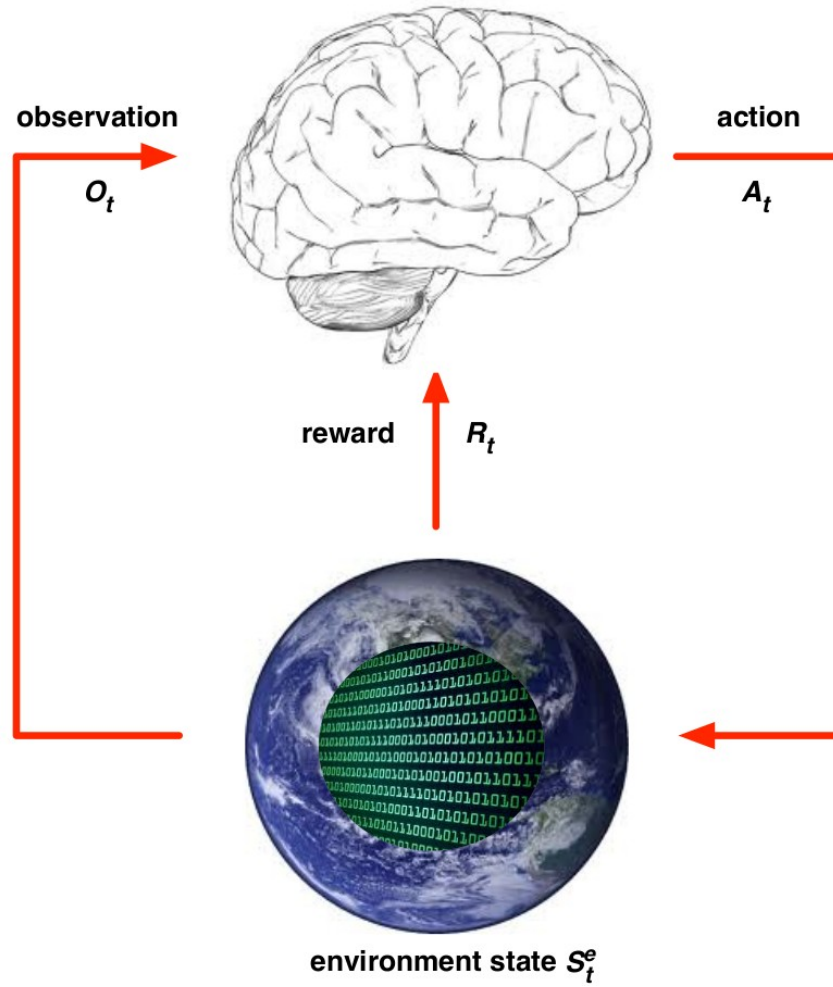
action

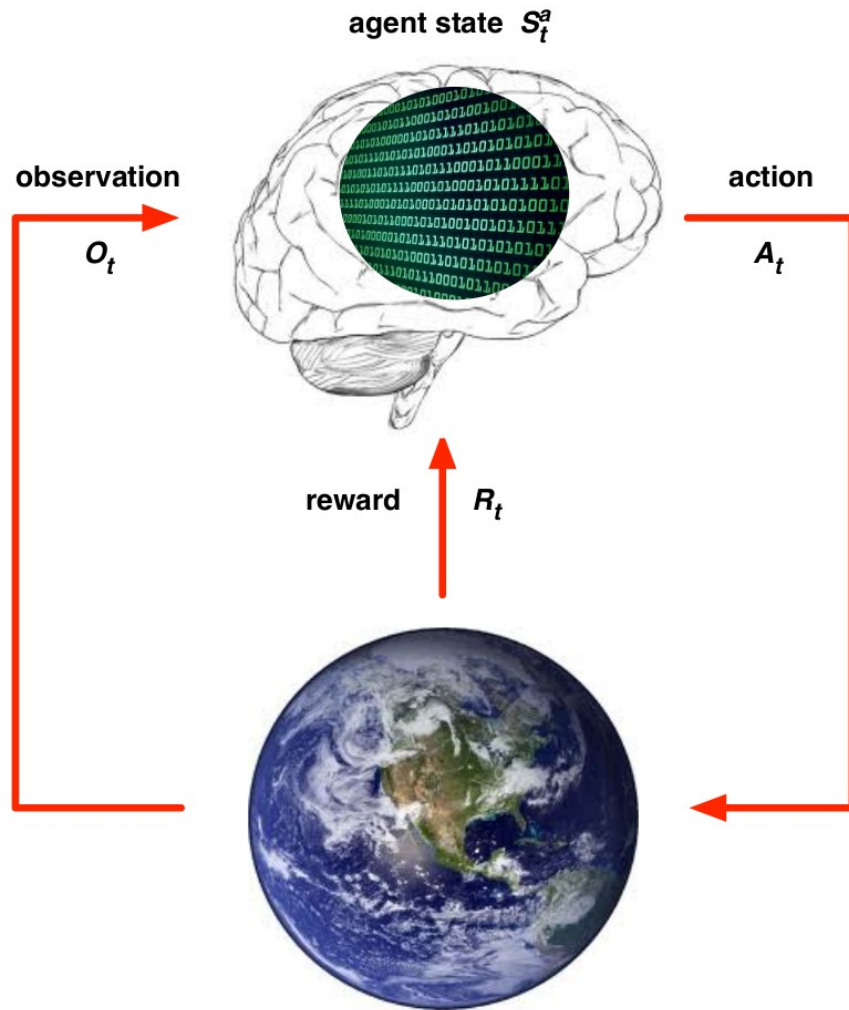
A_t

reward

R_t







Best food in the Campus/Town

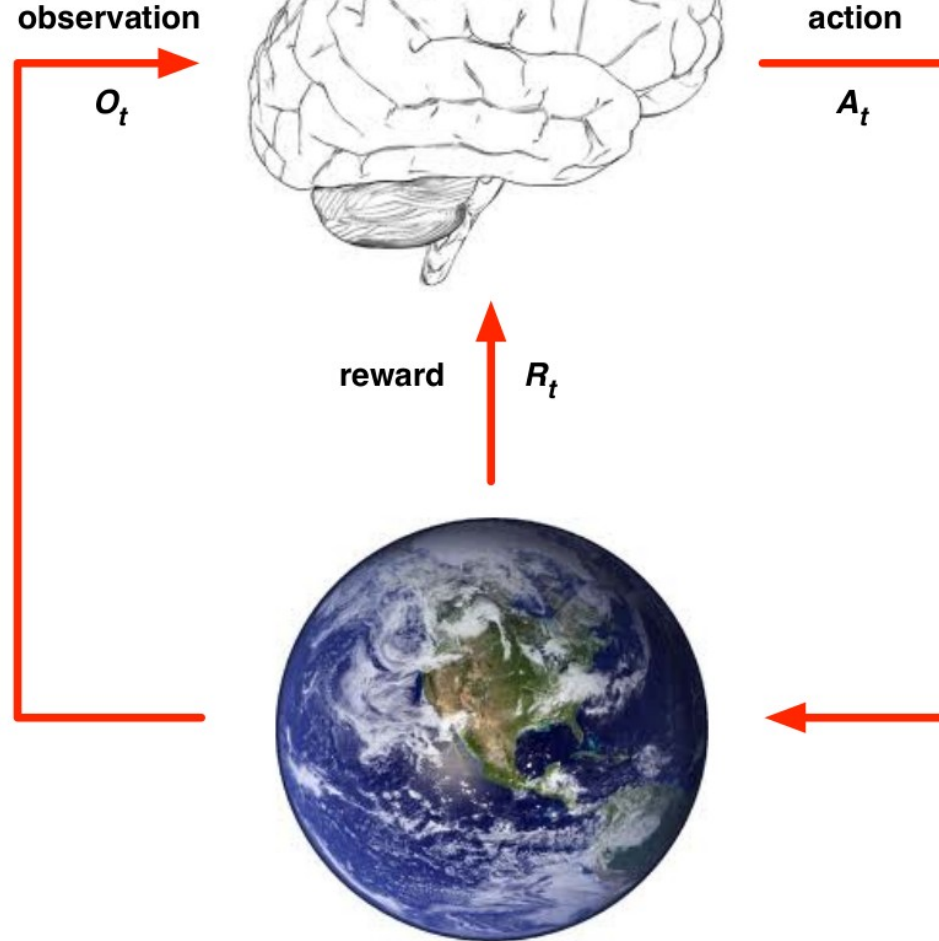
- Go to your favorite restaurant
- Try for a new restaurant in the campus/town

What is the deal?

- Exploration
- Exploitation

Robot motion planning

- Explore the environment
- Learn optimal or near-optimal movement



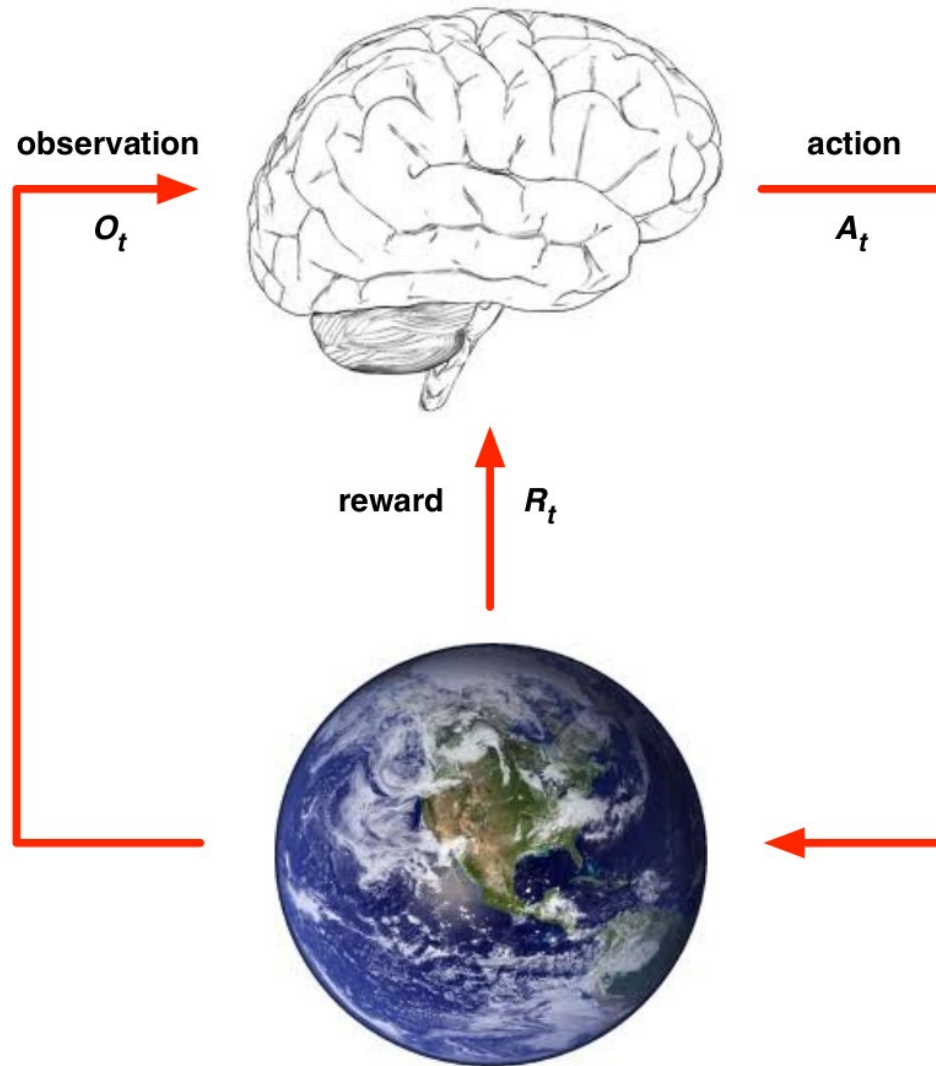
At each step t the agent:

- Executes action A_t
- Receives observation O_t
- Receives scalar reward R_t

The environment:

- Receives action A_t
- Emits observation O_{t+1}
- Emits scalar reward R_{t+1}

t increments at env. step



- **Agent**
- **Environment**
- **Actions**
- **States**
- **Rewards**

Autonomous vehicle

- Learn from the
 - Trial and error
 - Interaction with the environment
- Make decisions
 - Movement (Steering)
 - Pace (Acceleration and Braking)
- Maximize the reward function
 - Almost zero accident?
 - Fuel efficient?
 - Early arrival?
 - A combination of above?
 - Anything else?

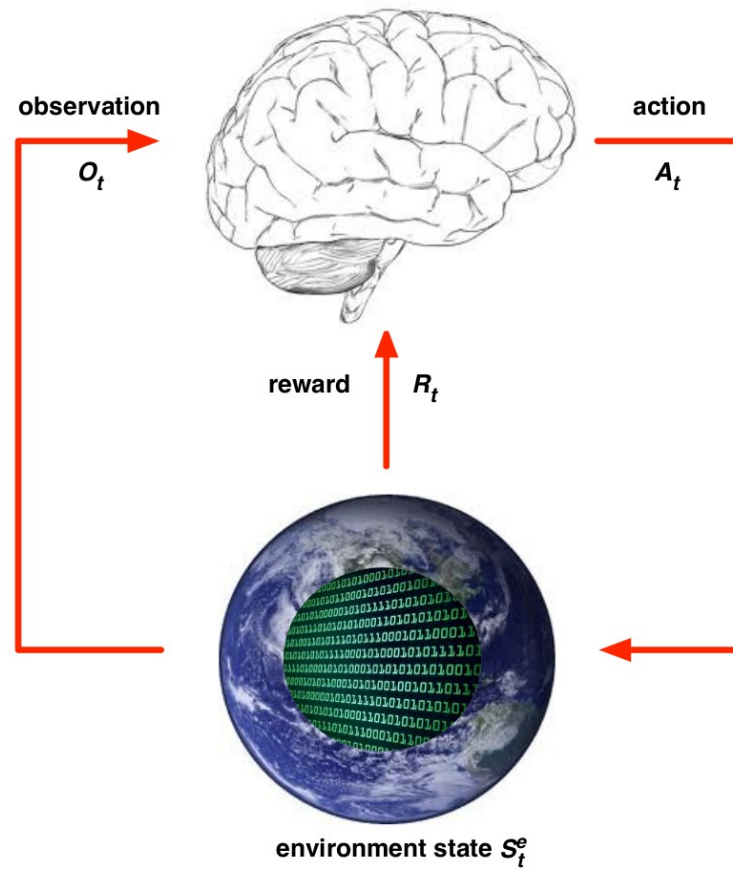
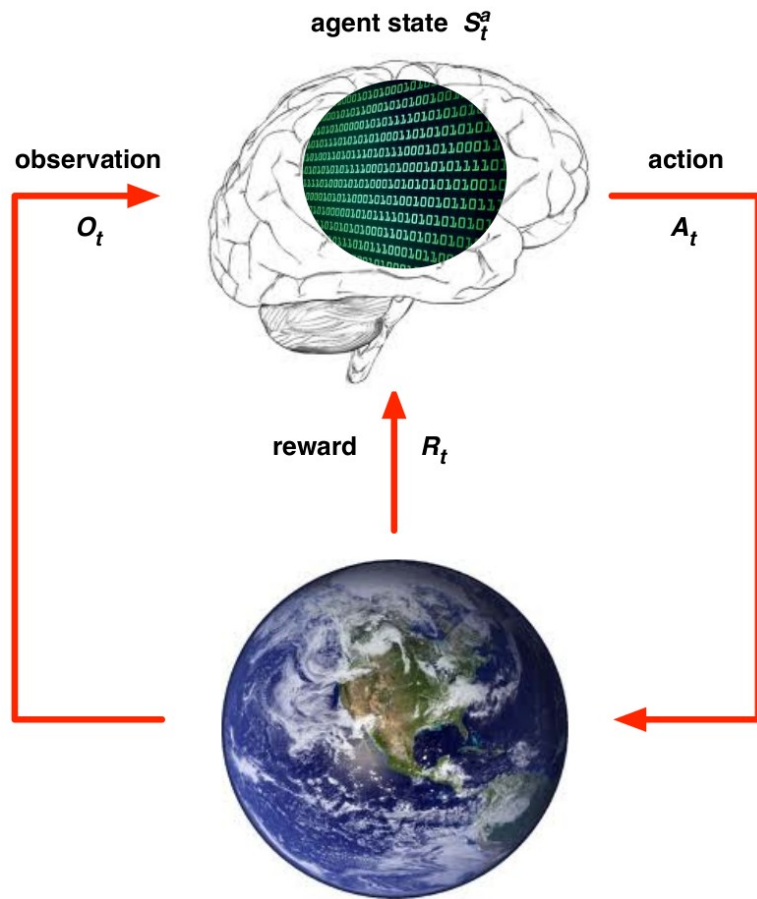
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Exploration vs. exploitation is a fundamental dilemma in reinforcement learning.

Healthcare

- Personalized treatment (dose control)
- Hospital management (resource management)
- Treatment recommendation (which therapy?)
- Molecular level application
- And many more ...



History and State

- The history is the sequence of observations, actions, rewards
$$H_t = O_1, R_1, A_1, \dots, A_{t-1}, O_t, R_t$$
- i.e. all observable variables up to time t
- i.e. the sensorimotor stream of a robot or embodied agent
- What happens next depends on the history:
 - The agent selects actions
 - The environment selects observations/rewards
- State is the information used to determine what happens next
- Formally, state is a function of the history:

$$S_t = f(H_t)$$

Information State

- An information state (a.k.a. Markov state) contains all useful information from the history.
- “The future is independent of the past given the present”
 - $H_{1:t} \rightarrow S_t \rightarrow H_{t+1:\infty}$
- Once the state is known, the history may be thrown away
- i.e. The state is a sufficient statistic of the future
- The environment state S_t is Markov
- The history H_t is Markov.

Stochastic Process

- A stochastic process is a collection of random variables indexed by time - it is a probability distribution over a space of paths; this path often describes the evolution of some random value, or system, over time.
- Example: - Simple Random Walk
 - Proposition
 - (i) $E[X_k] = 0$ for all k .
 - (ii) (Independent increment) For all $0 = k_0 \leq k_1 \leq \dots \leq k_r$, the random variables $X_{k_{i+1}} - X_{k_i}$ for $0 \leq i \leq r - 1$ are mutually independent.
 - (iii) (Stationary) For all $h \geq 1$ and $k \geq 0$, the distribution of $X_{k+h} - X_k$ is the same as the distribution of X_h .

Markov Chain

- The processes can be written as $\{X_0, X_1, X_2, \dots\}$, where X_t is the state at time t .
- On the transition diagram, X_t corresponds to where we are in at step t .
- If X_{t+1} depends only on X_t and it does not depend upon $X_0, X_1, X_2, \dots, X_{t-1}$. Then the process is called as Markov Chains.
- In a Markov chain, the future depends only upon the present: NOT upon the past.

Markov state

A state S_t is Markov if and only if

$$P[S_{t+1} | S_t] = P[S_{t+1} | S_1, \dots, S_t]$$

All the elements of a Markov chain model can be encoded in a transition probability matrix

$$A = \begin{pmatrix} p_{11} & p_{21} & \cdots & p_{m1} \\ p_{12} & p_{22} & \cdots & p_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ p_{1m} & p_{2m} & \cdots & p_{mm} \end{pmatrix}$$

$p_{ij} = P(X_{n+1} = j | X_n = i)$ $i, j \in S$. And the sum of each column is equal to one.

Markov Chain

Definition: The state of a Markov chain at time t is the value of X_t .

Definition: The state space of a Markov chain, S , is the set of values that each X_t can take.

Definition: A trajectory of a Markov chain is a particular set of values for $X_0, X_1, X_2, \dots$

Markov Property: only the most recent point in the trajectory affects what happens next.

Definition: Let $\{X_0, X_1, X_2, \dots\}$ be a sequence of discrete random variables. Then $\{X_0, X_1, X_2, \dots\}$ is a Markov chain if it satisfies the Markov property:

$$\mathbb{P}(X_{t+1} = s \mid X_t = s_t, \dots, X_0 = s_0) = \mathbb{P}(X_{t+1} = s \mid X_t = s_t),$$

for all $t = 1, 2, 3, \dots$ and for all states $s_0, s_1, \dots, s_t, s$.

Transition Matrix

Definition: Let $\{X_0, X_1, X_2, \dots\}$ be a Markov chain with state space S , where S has size N (possibly infinite). The transition probabilities of the Markov chain are

$$p_{ij} = P(X_{t+1} = j \mid X_t = i) \quad \text{for } i, j \in S, t = 0, 1, 2, \dots$$

Definition: The transition matrix of the Markov chain is $P = (p_{ij})$.

Transition Matrix is a square matrix.

Transition Matrix

Definition: The transition matrix of the Markov chain is $P = (p_{ij})$.

NOTE on Notation:

In the transition matrix P :

- the ROWS represent NOW, or *FROM* (X_t);
- the COLUMNS represent NEXT, or *TO* (X_{t+1});
- entry (i, j) is the *CONDITIONAL* probability that *NEXT* = j , given that *NOW* = i : the probability of going *FROM* state i *TO* state j .

$$p_{ij} = \mathbb{P}(X_{t+1} = j \mid X_t = i).$$

Choose

Question?

- The t-step transition probabilities

Let $\{X_0, X_1, X_2, \dots\}$ be a Markov chain with state space $S = \{1, 2, \dots, N\}$.

Recall that the elements of the transition matrix P are defined as:

$$(P)_{ij} = p_{ij} = \mathbb{P}(X_1 = j \mid X_0 = i) = \mathbb{P}(X_{n+1} = j \mid X_n = i) \quad \text{for any } n.$$

p_{ij} is the probability of making a transition FROM state i TO state j in a SINGLE step.

Question: what is the probability of making a transition from state i to state j over two/three/ t - steps?